

TRINH PHAN-CANH, MPharm. PhD.

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EDUCATION -----**Ph.D. in Molecular Genetics & Immunology**

Jan 2021 – Oct 2025

- *Medical University of Vienna – Max Perutz Labs Vienna, Vienna Biocenter Campus*
- Thesis: *Antifungal Multidrug Resistance and Virulence of Candida auris* (Grade: 1 – Excellence)
- Advisor: Prof. Dr. Karl Kuchler

M.Sc. in Pharmacy (Highest Distinction – Rector's Award)

Sep 2017 – Sep 2019

- *School of Pharmacy, University of Medicine and Pharmacy at Ho Chi Minh City (UMP)*
- Thesis: *Morphology, chemical composition and antimicrobial activity of Dirinaria applanata* (Grade: 10/10)

B.Sc. in Information Technology (Rector's Award for Academic Excellence)

Oct 2015 – Feb 2018

- *Vietnam National University – Ho Chi Minh City (VNU-HCM)*
- Focus: Management Information Systems (Distance Education Program)

National Pharmacist Degree (Rector's Award for Academic & Extracurricular Excellence) 2010 – 2015

- *School of Pharmacy, University of Medicine and Pharmacy at Ho Chi Minh City (UMP)*
- Thesis: *Fermentation conditions for Bacillus subtilis KP3 spores' production* (Rank 1; Grade: 10/10)

AWARDS AND DISTINCTIONS -----

- **Österreichische Mikrobiologie-Preis 2026 (Austrian Microbiology Prize 2026)** – ÖGHMP (2026)
- **Bernd Rode Award** – ASEA-UNINET Junior Researcher Award (2024)
- **Youth Travel Award** – FEBS Advanced Lecture Course, France (2024)
- **ESCMID Research Grant** – Outstanding project in fungal pathogenesis (2023)
- **FEMS Travel Grant** – Candida and Candidiasis Meeting, Canada (2023)
- **Ernst Mach Grant** – PhD Fellowship, ASEA-UNINET (2021–2023)
- **Pilot International Research Grant** – TRAC Sweden-Vietnam (2019)
- **Rector's Award, Master's Degree of Pharmacy** – UMP Vietnam (2019)
- **Youth Leaders Award** – Rector's Board, UMP Vietnam (2017–2019)
- **Homtamin Scholarship** – Korea United Pharm – Awarded for academic excellence (2015)
- **Daewoong Scholarship** – Daewoong Pharmaceutical – Awarded for academic excellence (2013)
- **First Prize – Scientific Research Contest: Seeking Elixir** – UMP (2012)
- **Student of Five Merits awards** – UMP (2011–2015)
- **Honorable Mention** – Thua Thien Hue Biology Olympiad (2010)
- **Thua Thien Hue Innovation Contest for Youth** – Awarded for developing biology learning app (2009)

PUBLICATIONS -----

1. **Phan-Canh T**, Penninger P, Seiser S, et al, Elbe-Bürger A & Kuchler K. *The carbonic anhydrase Nce103 controls fungal fitness and skin tropism of Candida auris*. **Nature Microbiology**, 2026. <https://doi.org/10.1038/s41564-025-02189-z>
This research resolves a 15-year question on intrinsic amphotericin B resistance.
Cover Image of Issue February 2026: <https://www.nature.com/nmicrobiol/volumes/11/issues/2>
Spotlight in Trends in Microbiology: <https://doi.org/10.1016/j.tim.2026.02.009>
2. **Phan-Canh T**, Jenull S, Bitencourt, et al. & Kuchler K. *White-Brown switching controls phenotypic plasticity and virulence of Candida auris*. **Cell Reports**, 2025, 115976. <https://doi.org/10.1016/j.celrep.2025.115976>
The first comprehensive molecular basis of epigenetic switching in *Candida auris*.

3. **Phan-Canh T**, Nguyen MLD, Phuc-Loi L, et al. & Kuchler K. *Rapid in vitro evolution of flucytosine resistance in Candida auris*. **mSphere**, 2025. <https://doi.org/10.1128/msphere.00977-24>
4. **Phan-Canh T**, Bitencourt T & Kuchler K. *Gene dosage of PDR16 modulates azole susceptibility in Candida auris*. **Microbiology Spectrum**, 2025. <https://doi.org/10.1128/spectrum.02659-24>
5. **Phan-Canh T** & Kuchler K. *Do morphogenetic switching and intraspecies variation enhance virulence of Candida auris?* **PLOS Pathogens**, 2024. <https://doi.org/10.1371/journal.ppat.1012559>
Featured on MicrobeTV Podcast: https://www.youtube.com/watch?v=DcHq_R8JaQM&t=1752s
6. **Phan-Canh T**, et al. & Kuchler K. *The cytochrome bc1 catalytic subunit Rip1 controls antifungal tolerance in Candida auris*. **Under revision, ACS Infectious Diseases**, 2026.
7. **Phan-Canh T**, Kuchler K., et al. *Pathogenicity and Virulence of Candida auris*. **Under revision, Virulence**, 2026.
8. **Phan-Canh T**, et al, Chauhan N & Kuchler K. *Antifungal resistance can emerge in Candida auris without drug selection pressure*. **Under revision, Microbiology Spectrum**, 2026.
9. Seiser S, Brezovec H, Penninger P, **Phan-Canh T**, et al., Kuchler K &, Elbe-Bürger A. *Human keratinocytes and fibroblasts coordinate early cutaneous innate defense against Candida auris*. **Under revision, Journal of Investigative Dermatology**, 2026.
10. Bitencourt T, Tsymala I, de Freitas Silva R, Wedige N, Prakash C, **Phan-Canh T**, et al. & Kuchler K. *Candida–Klebsiella interactions rewire fungal morphogenesis to create a host environment favoring pathogen survival with enhanced tissue pathology*. **Under review, Nature Communication**, 2026.
11. Tsymala I, Penninger P, Nogueira F, Jenull S, **Phan-Canh T**, et al. & Kuchler K. *Innate lymphoid cells integrate sensing and plasticity to control fungal infections*. **Cell Reports**, 2026. <https://doi.org/10.1016/j.celrep.2026.117140>
12. Cerbu D, Seiser S, **Phan-Canh T**, et al, Kuchler K & Elbe-Bürger A. *Octenidine effectively reduces Candida auris colonisation on human skin*. **Scientific Reports**, 2025. <https://doi.org/10.1038/s41598-025-11914-x>
13. Kavaklıoğlu G, Podhornik A, Vcelkova T, Marjanovic M, Beck MA, **Phan-Canh T**, et al. *L1TD1 associates with ORF1p to promote LINE-1 retrotransposition*. **eLife**, 2025. <https://doi.org/10.7554/eLife.96850.3>
14. Penninger P, Brezovec H, Tsymala I, Teufel M, **Phan-Canh T**, et al. & Kuchler K. *HDAC1 fine-tunes Th17 polarization in fungal infections*. **Cell Reports**, 2024. <https://doi.org/10.1016/j.celrep.2024.114993>
15. Bitencourt T, Nogueira F, Jenull S, **Phan-Canh T**, et al. Kuchler K, Lion T. *Multiomics reveals interspecies interaction pathways between A. fumigatus and K. pneumoniae*. **Communications Biology**, 2024. <https://doi.org/10.1038/s42003-024-07145-x>
16. Seiser S, Arzani H, Ayub T, **Phan-Canh T**, et al. & Kuchler K, Elbe-Bürger A. *Native human and mouse skin models to study Candida auris infection*. **Microbes and Infection**, 2023. <https://doi.org/10.1016/j.micinf.2023.105234>
17. Jenull S, Shivarathri R, Tsymala I, Penninger P, **Phan-Canh T**, et al. & Kuchler K. *Transcriptomics defines echinocandin resistance in Candida auris*. **mBio**, 2022. <https://doi.org/10.1128/mbio.00799-22>
18. Shivarathri R, Chauhan M, Mazumdar R, **Phan-Canh T**, et al. & Kuchler K & Chauhan N. *A proteomic approach for the quantification of post-translational protein lysine acetylation in Candida albicans*. In: *Candida species*. **Methods in Molecular Biology**, 2022. https://doi.org/10.1007/978-1-0716-2549-1_3

OUTREACH PUBLICATIONS -----

1. Phan-Canh T, Elbe-Bürger A & Kuchler K. *A Breath to Persist. CO₂ Fixation in Candida auris*, **SPECTRUM Dermatologie**, 2/2026.

ABSTRACTS, POSTERS, AND PRESENTATIONS -----

- **Talk** – ÖGMM Symposium 2026, im Rahmen des ÖIK, Saalfelden, Austria (Mar 2026)
- **Talk** – Mycology Meeting, Institute of Hygiene and Medical Microbiology, Innsbruck, Austria (Jan 2026)
- **Poster & Elevator Talk** – *Candida and Candidiasis*, Microbiology Society, Berlin, Germany (Oct 2025)

- **Poster & Poster Slam** – *Annual YSA PhD Symposium*, Medical University of Vienna (May 2025)
- **Poster** – *Host–pathogen interactions and virulence in human fungal pathogens*, FEBS Advanced Lecture Course, France (May 2024)
- **Poster** – *Candida and Candidiasis*, Microbiology Society, Montreal, Canada (Mar 2023)
- **Oral** – *Immune Control of Tissue Homeostasis*, TissueHome Symposium, Vienna (Jan 2023)
- **Poster** – VBC PhD Symposium, Vienna Biocenter Campus (Nov 2022)
- **Oral** – *Scientific Seminar of Young Vietnamese Intellectuals*, Prague, Czech Republic (Sep 2022)
- **Poster** – *Host–pathogen interactions and virulence in human fungal pathogens*, FEBS Advanced Lecture Course, France (May 2022)

PROFESSIONAL EXPERIENCE -----

Postdoctoral Researcher, BIDMC, Harvard Medical School from Jun 2026

- Development of spatial multi-omics technologies for host–disease interactions

Short-term Visiting Postdoc, Institute of Hygiene and Medical Microbiology, Innsbruck, Austria 2026

- Trained PhD students and staff in genetic engineering and bioinformatics
- Led a research project in antifungal drug discovery

PhD Candidate, Karl Kuchler Lab, Max Perutz Labs, Vienna Biocenter 2021–2025

- CRISPR-based genetic manipulation, multi-omics analysis (WGS, RNA-seq, proteomics, lipidomics)
- Host-pathogen and interkingdom interactions (yeast, *in vitro* with immune cells, *in vivo* mouse models)
- Drug resistance, microbial immunology and fungal pathogenesis

Lecturer, School of Pharmacy, UMP, Vietnam 2016–2021

- Led antimicrobial discovery projects using Myxobacteria and Streptomyces
- Supervised undergraduates and coordinated the Mycology Laboratory

Pharmacist & System Administrator, Drug Interactions Checker (thongtinthuoc.com.vn) 2015–2017

- Built and maintained drug info database; led safety content review and UX

Pharmaceutical biotechnology lab, UMP, Vietnam 2013–2015

- Undergraduate Research Assistant – Ho Chi Minh, Vietnam
- Investigation of sporulation dynamics in *Bacillus subtilis* during large-scale production; Mathematic models to optimize sporulation production of *B. subtilis*.

OTHER TECHNICAL SKILLS -----

- Wet lab techniques: Cloning, flow cytometry, biochemical assays, mouse models, microscope...
- Programming & Data Analysis: Proficient in R; working knowledge of Python, Bash, SQL...
- Omics Analysis: WGS, RNA-seq, proteomics, lipidomics (design, processing, interpretation); basic knowledge of scRNA-seq and spatial transcriptomics
- Visualization: ggplot2, GraphPad Prism, FlowJo, Adobe Illustrator, Photoshop, InDesign...
- Languages: English (professional fluency), Vietnamese (native)

REVIEWING EXPERIENCE -----

Editor-in-Training, International Journal of Antimicrobial Agents

Regular **Reviewer** for *ASM Microbiology Spectrum*, *Pathogens and Disease*, *Fungal Genetics and Biology*...

MENTORING & TEACHING -----

- Supervision and Mentorship of Students (Master's, Diploma, Internships)
Successfully supervised multiple students whose research projects received national awards:
 - **Best Diploma Thesis Award (High School Level), Austria** 2023
Mentees: Maja Prettnner and Katja Walter
 - **National Eureka Award research competition, Vietnam** 2019
Mentee: Duong Nguyen Phi Long
 - **Finding Elixir Competition at UMP, Vietnam** 2018
Mentees: Tran Quoc Vuong, Nguyen Xuan Loc, Nguyen Thi Loan
 - **Summer Research Program at UMP, Vietnam** 2017, 2018, 2019
Mentees: Vo Linh Tu, Pham Tran Kim Quyen, Tran Thi Thu Ha
- Developed curriculum and led student research groups at UMP 2016–2021

LEADERSHIP & OUTREACH -----

- Founder, PhD Talk: Seminar series for Vietnamese biologists in Austria - phdtalk.github.io 2021-2022
- President, Vietnamese Student Association in Austria 2022–2024
- Organizer of national and university science conferences 2020–2021
- Member, Executive Committee, Vietnamese Youth and Student Federation in Europe 2021–2024
- Editor, Pharmacy Brief News: A bilingual outreach publication 2020
- Founder, President, Pharmacy Soft Skill Club 2011–2015

REFERENCES -----

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